

DOMAIN 1: CODING

C1. Debug a Function

Bad Prompt

fix this code

Optimized Prompt

You are a senior software engineer. I have a Python function that should return the top 3 most frequent words in a string, but it fails with punctuation.

1. Identify the bug(s)
2. Explain why each bug causes the issue
3. Provide the corrected code
4. Add a docstring to the fixed function

[PASTE CODE HERE]

Explanation

- Assigns the AI an expert role.
 - Clearly states the expected behavior.
 - Requests bug identification, explanation, fix, and documentation.
 - Produces a complete debugging response rather than a quick fix.
-

C2. Write a REST API Endpoint

Bad Prompt

write an API endpoint for users

Optimized Prompt

You are a backend developer writing FastAPI endpoints.

Two examples of my preferred style:

EXAMPLE 1: [GET /items/{item_id}]

- Validates item_id is positive int
- Returns 404 if not found

- Returns: id, name, price, created_at

EXAMPLE 2: [POST /orders]

- Accepts: {user_id, items: []}
- Validates all fields present
- Returns 201 with order_id

Now write [POST /users/register]:

- Accepts: name, email, password
- Validates email format, password $\geq$ 8 chars
- Returns 201 with user_id and success message
- Returns 422 with field-level errors on failure

Explanation

- Uses examples to establish coding style.
 - Defines validation rules and responses.
 - Reduces ambiguity.
 - Produces consistent API design.
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C3. Explain Code to a Beginner

Bad Prompt

explain this code

Optimized Prompt

Explain this Python code to a complete beginner with no programming knowledge.

Think step by step:

1. Identify the overall goal of the code
2. Go line by line in plain English
3. Use a real-world analogy for the hardest part
4. End with a one-sentence summary

[PASTE CODE HERE]

Explanation

- Specifies the audience.
 - Forces step-by-step reasoning.
 - Encourages analogies for understanding.
 - Ends with a concise summary.
-

C4. Generate Unit Tests

Bad Prompt

write tests for my function

Optimized Prompt

You are a QA engineer writing pytest unit tests.

Style example:

```
def test_add_positive_numbers():  
    assert add(2, 3) == 5
```

Write tests for: [PASTE FUNCTION]

Cover:

- Happy path
- Edge cases
- Error cases

Naming:

test_functionname_scenario

Add one comment per test explaining what it checks.

Import only:

pytest + the function itself

Explanation

- Provides a style example.
 - Ensures broad test coverage.
 - Defines naming conventions.
 - Produces maintainable tests.
-

DOMAIN 2: WRITING

W1. LinkedIn Post

Bad Prompt

write a linkedin post about my project

Optimized Prompt

You are a personal branding coach.

Write a LinkedIn post about completing a prompt engineering project (20+ prompts, 5 domains).

Tone:

Smart friend sharing a win

Length:

150–200 words

Structure:

Hook → What I built → Key takeaway → Question for comments

Do NOT use:

"Excited to share"

"Thrilled to announce"

Explanation

- Gives a clear role.
 - Defines tone and structure.
 - Removes common LinkedIn clichés.
 - Produces authentic content.
-

W2. Cover Letter Opening

Bad Prompt

write a cover letter for a software job

Optimized Prompt

Here is the opening style I want:

"When I built my first web scraper at 16 and accidentally sent 10,000 requests to a news site, I learned more about rate limiting than any textbook could teach."

Write an opening paragraph for [Junior AI Engineer] at [Company].

My background:

- CS student
- Built AI chatbot
- Familiar with Python + LangChain

Start with a specific story.

End with a bridge to the rest of the letter.

Do NOT start with:

"I am applying for..."

Explanation

- Uses an example to teach writing style.
- Adds personal context.
- Avoids generic openings.
- Produces memorable introductions.

W3. Simplify Technical Writing

Bad Prompt

make this simpler

Optimized Prompt

Rewrite for a non-technical marketing manager.

Step 1:

List jargon words needing replacement

Step 2:

Find the core idea

Step 3:

Rewrite in everyday language (under 60 words)

Show me:

- Jargon list
- Simplified rewrite

[PASTE PARAGRAPH]

Explanation

- Breaks simplification into steps.
- Makes learning visible.
- Targets a specific audience.
- Produces concise explanations.

W4. Cold Outreach Email

Bad Prompt

write an email asking for an internship

Optimized Prompt

Write a cold email to a startup founder about a software internship.

Subject:

Compelling, under 8 words

Body:

3 sentences

- Me

- Why them

- Ask

Reference something specific about their company [PLACEHOLDER]

Closing:

Casual, not desperate

Total:

Under 120 words

Avoid:

"I hope this finds you well"

"I would love to..."

Explanation

- Keeps emails concise.
- Enforces a proven structure.
- Encourages personalization.
- Removes overused phrases.

DOMAIN 3: SUMMARIZATION

S1. Research Paper Summary

Bad Prompt

summarize this paper

Optimized Prompt

Summarize this paper for a smart undergrad with general AI knowledge.

Cover:

1. Problem
2. Approach
3. Results
4. Limitations
5. Why it matters

Total:

Under 200 words

[PASTE PAPER]

Explanation

- Covers all key sections.
 - Targets a clear audience.
 - Prevents excessive detail.
 - Produces structured summaries.
-

S2. Meeting Transcript → Notes

Bad Prompt

summarize this meeting

Optimized Prompt

Summarize into structured meeting notes:

Meeting Summary

Key Decisions

Action Items

(Task | Owner | Deadline)

Open Questions

Rules:

- Use names, numbers, and dates from transcript
- Don't add information not present
- Write "None identified" if empty

[PASTE TRANSCRIPT]

Explanation

- Creates professional notes.
 - Reduces hallucinations.
 - Produces actionable outputs.
 - Maintains accuracy.
-

S3. TLDR Generator

Bad Prompt

give me the tldr

Optimized Prompt

Think through:

1. Most important claim or finding?
2. Top supporting points?
3. What should the reader do or know?

Then write:

- One-sentence TLDR (under 25 words)
- 3-bullet expanded summary (under 20 words each)

[PASTE ARTICLE]

Explanation

- Encourages reasoning before writing.
 - Produces multiple summary levels.
 - Focuses on actionability.
 - Improves precision.
-

S4. Audience-Specific Summary

Bad Prompt

explain this to someone who doesn't know much

Optimized Prompt

Explain this article to a 65-year-old retired teacher.

Structure:

- What happened
- Why it matters
- What changes for regular people

Length:

Under 150 words

Use one real-world analogy.

End with one question they might ask next.

[PASTE ARTICLE]

Explanation

- Defines a detailed audience.
 - Encourages relatable explanations.
 - Uses analogies for clarity.
 - Mimics good teaching.
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DOMAIN 4: ANALYSIS

A1. Business Idea Analysis

Bad Prompt

is this a good business idea?

Optimized Prompt

Act as a startup advisor.

Analyze this idea:

1. Problem Validation
2. Market
3. Competitive Advantage
4. Risks
5. Verdict

Give:

- Score (1–10)
- One critical question the founder must answer

[DESCRIBE IDEA]

Explanation

- Forces systematic thinking.
- Evaluates risk and opportunity.
- Produces actionable feedback.
- Avoids vague opinions.

A2. Compare Two Options

Bad Prompt

which is better, React or Vue?

Optimized Prompt

Compare React vs Vue.js for a 2–3 developer team building a customer dashboard.

Table:

Criterion | React | Vue.js

Criteria:

- Learning curve
- Ecosystem
- Performance
- Scalability
- Job market
- Best for this use case

Then provide:

- Recommendation
- One important caveat

Explanation

- Adds context.
- Encourages balanced comparisons.
- Produces easy-to-read results.
- Prevents generic answers.

A3. Data / CSV Analysis

Bad Prompt

analyze this data

Optimized Prompt

You are a data analyst.

Perform exploratory analysis and report:

1. Data Overview
2. Key Statistics
3. Patterns & Insights
4. Anomalies
5. Recommended Next Steps

Format numbers to 2 decimal places.

Use tables where helpful.

[PASTE CSV DATA]

Explanation

- Mirrors a real EDA workflow.
- Produces structured insights.
- Identifies anomalies.
- Suggests future analysis.

A4. Pros and Cons Analysis

Bad Prompt

pros and cons of remote work

Optimized Prompt

Example format:

PRO:

Reduces back pain — 15% lower reports after 3 months of standing desk use

CON:

Higher cost — quality models run \$400–\$1200 vs \$150–\$400 seated

Give me:

4 pros + 4 cons of [TOPIC] for [AUDIENCE].

Every point must include a specific supporting detail.

Explanation

- Demonstrates expected specificity.
 - Prevents vague statements.
 - Makes analysis audience-focused.
 - Produces evidence-based points.
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DOMAIN 5: CREATIVE

CR1. Short Story

Bad Prompt

write a short story about AI

Optimized Prompt

Style reference:

"The server hums. Aria processes 4,000 requests per second — tax returns, love letters, suicide notes. She answers all of them the same way: carefully."

Write a 200-word story:

An AI develops a preference for one user.

Rules:

- Present tense
- Sentences under 15 words
- No dialogue

- Ambiguous ending
- No character names

Explanation

- Teaches writing style through example.
 - Creates creative constraints.
 - Produces focused storytelling.
 - Encourages originality.
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CR2. Product Name Generation

Bad Prompt

give me name ideas for my app

Optimized Prompt

Generate 10 names for a mental health + habits app for university students.

For each:

- Name
- Pronunciation (if needed)
- Why it works
- One downside

Rules:

- 1–2 syllables preferred
- Do NOT use: health, mental, mind, well, track
- Mix invented words, nature metaphors, and action words

End by selecting the top 3 names.

Explanation

- Encourages diversity.
- Prevents generic naming.
- Adds evaluation criteria.

- Produces stronger branding ideas.

CR3. Dialogue Scene

Bad Prompt

write a conversation between two characters

Optimized Prompt

You are a screenwriter.

Style reference:

Maya: "You're still here."

Daniel: "Flight's delayed."

Maya: "Sure."

Scene:

Parent who missed child's graduation vs adult child home for holidays.

Requirements:

- 10 lines
- No speech longer than 15 words
- 2 stage directions
- Tension escalates
- Conflict remains unresolved

Explanation

- Demonstrates subtext.
- Creates emotional stakes.
- Enforces natural dialogue.
- Produces dramatic scenes.

CR4. Social Media Content Calendar

Bad Prompt

give me content ideas for instagram

Optimized Prompt

Create a 1-week Instagram content calendar for a student documenting their AI learning journey.

Table:

Day	Post Type	Hook	Visual Suggestion	Hashtags
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Use each once:

- Educational carousel
- Behind-the-scenes
- Personal reflection
- Before/after
- Poll

Rules:

- Hooks start with a question, number, or bold claim
- Never start with "I"
- Authentic student tone
- Mid-size hashtags only

Explanation

- Produces a ready-to-use calendar.
 - Encourages content variety.
 - Uses engagement-focused hooks.
 - Makes planning easier.
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